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Wavelength scanning and mirror alignment of a new multipass Fabry-Perot interferometer

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Abstract A new arrangement for five-pass interferometry is presented together with a holder for wavelength scanning and mirror alignment of a multipass Fabry-Perot interferometer. The parallelism of the mirror surfaces can be controlled precisely through independent rotations around two mutually perpendicular axes and the distance between the mirrors can be varied by 860 nm independently of the angular controls. The operation of the holder has been tested using a Tolansky interferometric arrangement. Necessary modifications in the design for scanning in other wavelength ranges are outlined.

1 Introduction

Fabry-Perot interferometers are frequently used for studying the Brillouin components of light scattered from solids. When in use it is important that (1) the two mirrors of the interferometer should be kept parallel to each other to within, say, 5 nm over an area of 25 mm², i.e. to within $\pm 1 \times 10^{-6}$ rad; (2) it should be possible to vary the distance between the mirrors so that several interference orders pass one point in the pattern and thus make it possible to scan the spectrum of the scattered light. These two problems have been tackled differently by various workers. To keep the two mirrors parallel one standard technique is to put quartz spacers between them (Brochard *et al* 1952, Jacquinet 1960). However, with this method one is strictly limited to the use of pressure scanning which becomes increasingly difficult to apply when working with large free spectral ranges or small mirror spacings (Gustafsson *et al* 1974a, b, Rockwell *et al* 1976, Rank and Shearer 1956). Three micrometer screws in series with three piezoelectric transducers for mirror alignment and a sawtooth voltage applied to the transducers for scanning have also been used (Sandercock 1970). For large mirror spacing adjustments, in the range of 1–2 mm, a holder with three flat metal springs has been used (Greenler 1959). This is capable of keeping the mirrors parallel to within about 5×10^{-6} rad by manipulating a micrometer screw attached to one of the springs after the mirrors have been moved.

In our design, we use elastic rotations of solid bridges driven by piezoelectric elements to achieve both mirror alignment and scanning. The alignment is obtained by

changing the angle between the mirror surfaces through independent rotations of two mutually perpendicular bridges. The distance between the mirrors can be varied by about 860 nm with the holder designed by us without affecting the alignment as opposed to other mechanical designs, e.g. that of Greenler, in which mirrors have to be aligned again after the movement.

The most important design criteria of this new mechanical device are the dimensions of the solid bridges and the different stiffnesses against rotation around perpendicular axes. By a proper choice of parameters, it is possible to produce a very precise bending of the bridge around one axis.

To achieve the translation of one of the Fabry-Perot mirrors, we used four solid bridges placed at the corners of a parallelogram. By applying a force parallel to two of the sides of the parallelogram, a minute bending of all four bridges occurred causing a small translational movement of one side of the parallelogram with respect to the opposite one. By changing the lengths of the sides and the point of application of the force, the maximum distance of translation can be varied considerably.

The holder presented here has been designed for a new five-pass Fabry-Perot interferometer recently developed at Chalmers University of Technology. This interferometer uses the hypotenuse faces of the prisms coated with dielectric multilayers to act as mirrors (figure 1) and gives the same

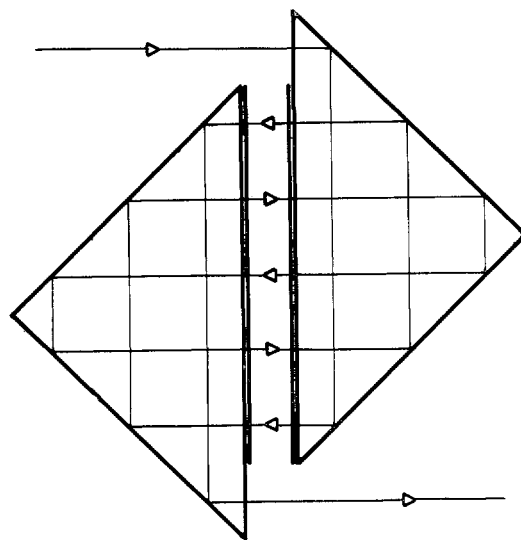


Figure 1 Ray sketch of five-pass Fabry-Perot interferometer.

contrast but lower intensity losses than those commercially available at present. By using this simple prismatic arrangement it is possible to design a very compact instrument and also to use the mirror surfaces very efficiently compared with the conventional arrangement of Fabry-Perot interferometer and corner prisms.

2 Design of the holder

Figure 2 shows a sketch of the holder. Essentially, it consists of two platforms (1, 5) to support the two Fabry-Perot elements (7). When aligning the surfaces parallel to each other, the mirror mounted on the left-hand platform is held fixed. The one mounted on the right-hand base plate is aligned with the help of elastic rotations about two mutually perpendicular bridges. These bridges, together with the piezoelectric elements used to drive them, are shown in figure 2(b).

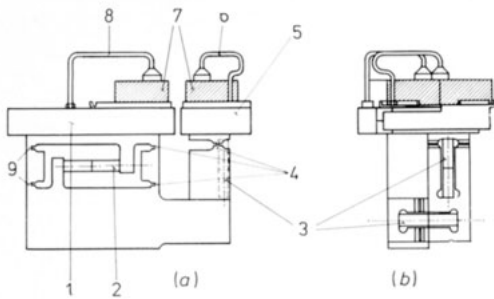


Figure 2 (a) Front view, (b) side view of the holder. (1, 5) platforms, (2, 3) piezoelectric elements, (4, 9) solid bridges, (6, 8) press rods to keep the Fabry-Perot prisms (7) in position.

A bridge consists of a block of material, in our case construction steel, which is weakened at one place by the cutting of a slot. This forms a 'bridge' of length l , width b and thickness h (figure 3). Near the bridge, space is provided to fit the piezoelectric elements. The elements (type PXE-21, AB ALCOMA, Philips), driven by electrostatic voltages, exert a force causing elastic rotations around the desired axis e.g. the z axis in figure 3.

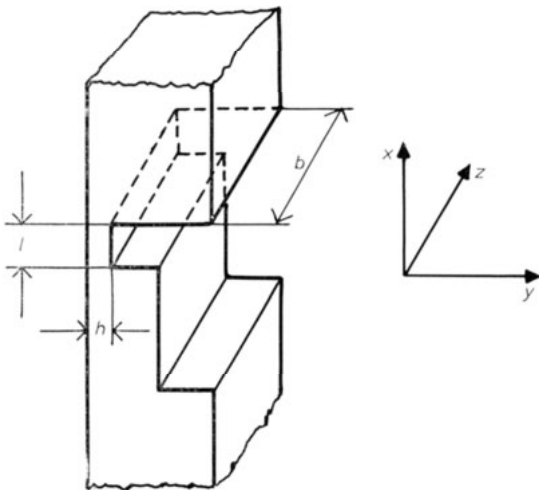


Figure 3 A solid bridge with width along the z axis.

Elementary calculations on bridge stiffness and a knowledge of the elastic properties of the construction material show that, for a bridge having its width along the z axis and the dimensions $l=0.8$ mm, $h=1.1$ mm, $b=23$ mm, the stiffness against rotation around the x axis is about 150 times, and around the y axis about 400 times as large as that around the z axis (appendix). This effectively checks the rotations of the bridge around the undesired axes.

Mechanical disturbances may be caused by vibrations. To minimise the influence of vibrations, the resonance frequencies of either part of the holder, consisting of the two platforms, should be kept reasonably high. The resonance frequencies of the two parts were experimentally measured and found to be higher than 1000 Hz for our particular construction (figure 2). The resonance frequency can most easily be changed by altering the mass of a platform.

For scanning, similar solid bridges are used in a parallelogram configuration. Four bridges form the corners of the

parallelogram as can be seen in figure 2(a). A force exerted by a piezoelectric element (2) slightly bends the bridges and thus the distance between the mirrors is varied. There is also a small movement perpendicular to the scanning direction. This is estimated to be 25 nm for a parallelogram having sides 20 and 60 mm long but, being in the plane of the mirrors, does not affect the optical measurements.

3 Experiments

The operation of the holder (figure 4) has been tested using Tolansky fringes between two optical flats placed on the two platforms. One flat was coated to give a highly reflective mirror

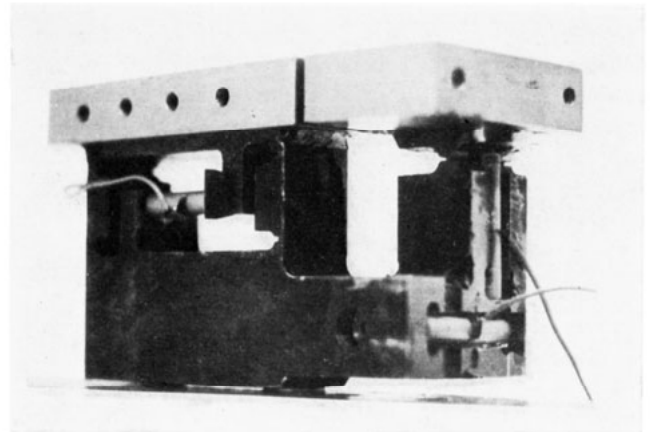


Figure 4 Photograph of the holder without the press rods.

while the other one, facing the beam splitter and the microscope, had a silver coating to give about 70% reflectivity. The fringes were produced by light from a 632.8 nm He-Ne laser. A microscope having a micrometer eyepiece was employed to measure the fringe positions.

When a voltage is applied to the piezoelectric elements that are vertical in figure 2(b), (i) the fringe spacing changes; (ii) the angle between the fringes changes; and (iii) the fringes move due to the change in the distance between the optical

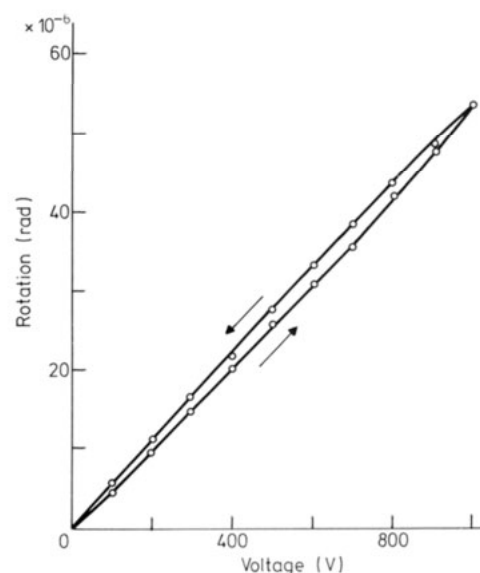


Figure 5 Angular adjustments with a total voltage change of 1000 V applied in steps to the vertical set of piezoelectric elements in figure 2(b).

flats. It proved that (iii) is the predominant effect and is most suitable for measuring rotations. This method determines the relative changes in the angles. Figure 5 shows the results of the measurements with a voltage change of 1000 V. The individual points were measured with an accuracy of $\pm 1 \times 10^{-6}$ rad. For the maximum voltage, 2000 V, the rotation is 136×10^{-6} rad. In addition, our investigation of the effect of very small voltage changes shows that rotations can be controlled in steps smaller than 0.5×10^{-6} rad. This is intelligible from the fact that the maximum voltage gave a rotation of 136×10^{-6} rad. Similar results were found for the vertical bridge in figure 2(b).

As the relative spacing of the optical flats is changed by applying a voltage to the piezoelectric elements (2) in figure 2(a), the fringes move past the cross-wire of the micrometer eyepiece. If one fringe passes the cross-wire, the distance scanned equals half the wavelength of the light used. The graphs in figure 6 show the scanning produced as the voltage

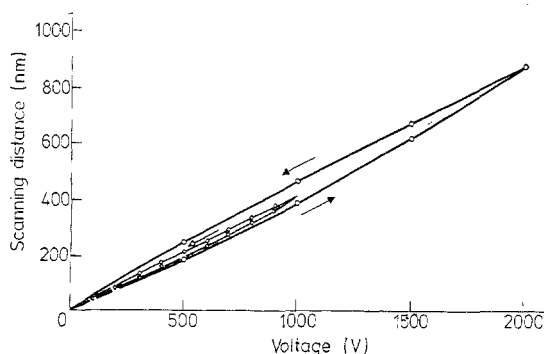


Figure 6 Scanning graphs.

on the piezoelectric elements is increased from zero to 2000 V in steps of 500 V and then back to zero. The smaller loop corresponds to 0–1000 V in steps of 100 V and then back to zero. In either case the lower part of the loop corresponds to increasing voltage. The measurements are accurate to $\pm 2\%$. The hysteresis effects are due to inherent properties of the ceramic of ferroelectric piezoelements. Since the hysteresis is comparatively small, it should not disturb the operation of the holder.

4 Conclusions

The experiments show that the holder is capable of aligning the two Fabry–Perot mirrors to better than 1×10^{-6} rad. The spacing between the mirrors can be varied over a convenient range when studying visible radiation. The number Δm of free spectral ranges, scanned in the scattered light spectrum of vacuum wavelength λ , is related to the index of refraction n of the gas surrounding the interferometer and the scanning distance Δd by $\Delta m = 2n\Delta d/\lambda$. This would imply a scanning of three free spectral ranges in the scattered light spectrum while using, say, an argon ion laser of 488 nm. This is a suitable scanning range for Brillouin scattering experiments. By modifying the design slightly so that the scanning piezoelectric elements are placed at one-fifth of the height of the scanning parallelogram, the scanning range can be increased by a factor of 5. The holder then can be used for scanning in the infrared and would cover two free spectral ranges if one used a He–Ne laser of 3391 nm. This holder can also be used for Raman scattering experiments where the distance between

the mirrors is much smaller, typically of the order of 0.15 mm, and it becomes necessary to use an additional Fabry–Perot interferometer (Rockwell *et al* 1975, Murray *et al* 1975, Woerner and Graytak 1976).

While this device is used manually, the temperature has to be kept constant to within $\pm 0.05^\circ\text{C}$ because of the different thermal properties of the construction material of our holder and the piezoelectric elements. With simple thermal insulation around the holder this is not very difficult to achieve for the rather short duration of the measurements. However, by using either a construction material having thermal properties close to those of the piezoelectric elements or an automatic thermal stabiliser (Sandercock 1970) the holder can be made practically completely insensitive to temperature variations in the laboratory.

Appendix

The stiffnesses of a bridge, having length l along the x axis, width b along the z axis, and thickness h , against rotations around the x , y and z axes are

$$C_x = \frac{1}{12} \frac{G}{l} (bh^3 + hb^3); \quad C_y = \frac{1}{12} \frac{E}{l} hb^3; \quad C_z = \frac{1}{12} \frac{E}{l} bh^3,$$

where G is the shear modulus and E the modulus of elasticity of the material.

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